

The Aromatic Garden

Exploring new ways to interactively interpret narratives combining olfaction and vision including temporal change of scents using olfactory display

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CCS CONCEPTS

• **Human-centered computing** → Interaction design.

KEYWORDS

Olfaction, Interaction

ACM Reference Format:

Saya Onai, Nathan Cohen, and Takamichi Nakamoto. 2021. The Aromatic Garden: Exploring new ways to interactively interpret narratives combining olfaction and vision including temporal change of scents using olfactory display. In *SIGGRAPH Asia 2021 Emerging Technologies (SA '21 Emerging Technologies)*, December 14–17, 2021, Tokyo, Japan. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3476122.3484840>

1 BACKGROUND

Visual and auditory human interface technologies have been developed and are utilized worldwide, but olfactory devices are not yet so common. Olfaction is a key human sense, with smells capable of inducing strong emotions and memory associations. Prior research engages with smell in a digital context for entertainment or hazard simulation, including games incorporating scent, governing moods [Amores, 2017], although with limitations in interaction or precision. Our sense of smell also supports our visual and auditory senses, which tend to be quicker and more accurate, to enable us to achieve a more complete sensory perception of our environment.

This research seeks to enable players of the game we are creating, titled ‘The Aromatic Garden’*, to navigate with greater ability through olfaction. To realize this, we have created a game with interactive emissions of scent and temporal controls. The visual content includes a fantastical and detail filled set of 4 related landscapes encouraging the viewer to use their imagination to explore them. 12 of the items depicted in these landscapes are associated with particular scents. Using these as sources, we have set the goal of our game to searching for and collecting scents that can be located within the 4 landscape images. To reinforce the olfactory emphasis we also limit the player’s visual information by covering the landscape images with tiles at the beginning of the game. As the player navigates, the tiles disappear momentarily, so they can simultaneously track their journey, enjoy revealing details of the

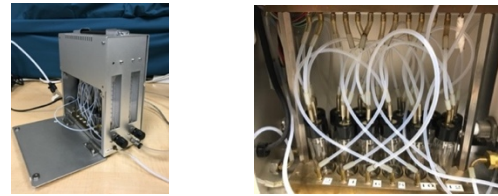


Figure 1: The olfactory display and scent containers

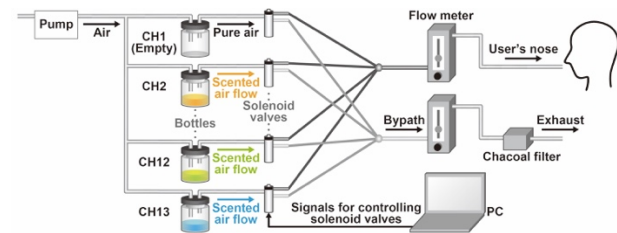


Figure 2: The structure of the olfactory display

landscape contents and concentrate on seeking and collecting the scents hidden within it.

2 CONTEXT

Ranasinghe et al. created a computer game with scents as cues [Ranasinghe et al., 2019]. Although they claimed that scents can be used for making stories into interactive experiences, their olfactory device was too simple to specify the mixture composition and to enable it to change according to time.

We have developed an olfactory display, based on solenoid valves, with the capability to control mixture composition precisely and to effect its temporal change as is shown in Figures 1 and 2 [Nakamoto et al., 2007]. When it emits a smell, the user starts to receive the scent in less than 10 seconds. The stimulus becomes weak within less than 1 minute after the display stops emission [Iseki et al., 2021]. Although we have already made a cooking game with scents [Nakamoto et al., 2008], this study is the first experience of making a story by utilizing the temporal change of mixture composition. Our olfactory display provides 12 different scents with the ability to mix them in varying percentages. By using this display, we can present 12 scents that also respond in intensity to the distance between the player and the scent source in the game and translate temporally. We believe that this enhanced capability of scent emission will provide a more dynamic and immersive user experience, and enable a deeper narrative association and interpretation.

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SA '21 Emerging Technologies, December 14–17, 2021, Tokyo, Japan

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ACM ISBN 978-1-4503-8685-2/21/12.

<https://doi.org/10.1145/3476122.3484840>

3 METHOD

3.1 The game abstract

This game, ‘The Aromatic Garden’, provides a journey into the 4 landscapes’ imagery guided by various scents. To enable the player to move between each landscape to collect all 12 scents, and thereby complete the first level of the game, we have created ‘scent tunnels’. Each tunnel has a pair of scents with one presented at each entrance and exit. When the player reaches one of the scent sources, they automatically enter the tunnel. After a certain length of time the tunnel ends and the player arrives at a different location in a different landscape from where they began. The pair of scents are collected together when the journey through each scent tunnel is completed.

3.2 Smell presentation

The principle form of smell expression in this game is the player searching for a new scent. The scent emissions change in intensity according to the distance between the player and the scent source. Initially, as the player navigates the tiled painting a weak scent might be detected growing in intensity as they approach the scent target. Only one scent target is active at a time. To devise this form of scent emission, we conducted a pilot test with 10 people to ascertain their sensitivity to scent type and intensity gradation. Based on the results, we selected the scents that were most readily recognizable when their intensities change gradually.

In relation to the scent tunnels paired scents are emitted as a temporally changing composition, with the player perceiving the first scent gradually changing to the second scent. To be sure that the player recognizes the change we conducted two more pilot experiments. The first test established the optimal scent pairs, and the second checked that the player could sense the change and how long the time for traveling in each tunnel should take to achieve this. In the first experiment, 10 participants were presented with 12 smells and asked to plot their impression of them into triangles with vertices representing ‘pleasant,’ ‘natural’ and ‘unpleasant’ (see Figure 3). To make the changes more easily recognizable we selected pairs of scents that have contrasting impressions. For example, ‘Fresh Cut Grass’ and ‘Cade Oil’ show different response tendencies in the Figure 3 plots, so we paired these scents. For the second experiment, participants experienced one of the tunnels of smells over 2 different time durations and conditions: 20 and 30seconds for the duration of the tunnel, and without and with visual information for the time spent in the tunnel. Given the results we have been able to determine that most participants do recognize the scent changes and the optimal time for this to occur will be over a duration of 25 seconds. During 25 seconds, the ratio of each scent changes linearly from 100% to 0%, and the other changes in reverse. The gradual change between paired scents with strong contrast tells an important piece of information to players that they are moving to a different place.

4 SOCIAL IMPACT

This research offers a dynamic introduction to olfaction in the context of gaming while utilizing imagery that provides for narrative interpretation. This combination of art and new olfactory

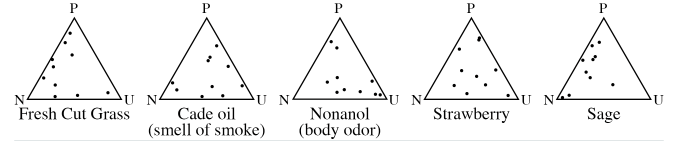


Figure 3: Triangles with plots (P, N and U represent pleasant, natural and unpleasant respectively)

Table 1: Goals for different levels of the Aromatic Garden game

Level	The goal for each level
Level 1	Searching for and collecting smells
Level 2	Relocating scents to their original locations
Level 3	Mixing 12 scents freely with corresponding parts of the picture reassembled as a new picture
Level 4	Scent comparison and pairing
Level 5	Detecting which smell is included in a given mixture of smells
Level 6+	Adaptation of process above using abstract (non-referential) image

technology is novel, with the use of olfaction as the leading sense presenting an engaging and challenging experience. We envisage that the people who play our game will be impressed by this experience in both its mental and physically engaging aspects, and that this could lead to further development of multi-sensory interactive technology, design and games. There is also the potential for this game to enable olfactory training toward enhancing smell perception, although this will form part of our future research.

5 FUTURE DEVELOPMENTS

To advance this project, we regard the initial goal of the ‘Aromatic Garden’ (collecting scents) as ‘Level 1’ of a number of levels for the game. Additional tasks will be explored through creating levels 2, 3 and so on, (see Table 1), with those at higher levels potentially including player input. We are also investigating how temporal shifts in scent presentation and perception could lead to new scent mixtures that have distinctive properties that would be usable for future applications. We intend for this to generate new research about human olfaction and cognition.

* Inspired by the painting, ‘The Garden of Earthly Delights’, Hieronymus Bosch (1490-1500)

ACKNOWLEDGMENTS

This work was partially supported by Tokyo Tech World Research Hub Initiative (WRHI) and KAKENHI under Grant JP21H04889.

REFERENCES

- Judith Amores, Essence: Olfactory Interfaces for Unconscious Influence of Mood and Cognitive Performance, CHI’17, 2017, 28-34.
- Nimesha Ranasinghe *et al.* 2019. Tainted: An olfaction-enhanced game narrative for smelling virtual ghosts. International Journal of Human-Computer Studies, Volume 125, May 2019, Pages 7-18.

Takamichi Nakamoto *et al.* 2007. Improvement of olfactory display using solenoid valves. *IEEE Virtual Reality*, 179-186.
Takamichi Nakamoto *et al.* 2008. Cooking Up an Interactive Olfactory Game Display,

IEEE CG and applications, pp. 75-78.
Masaaki Iseki *et al.* 2021. Evaluation of Temporal Characteristics of Olfactory Displays with Different Structures, *IEICE Trans.* vol.E104-A, 744-750.